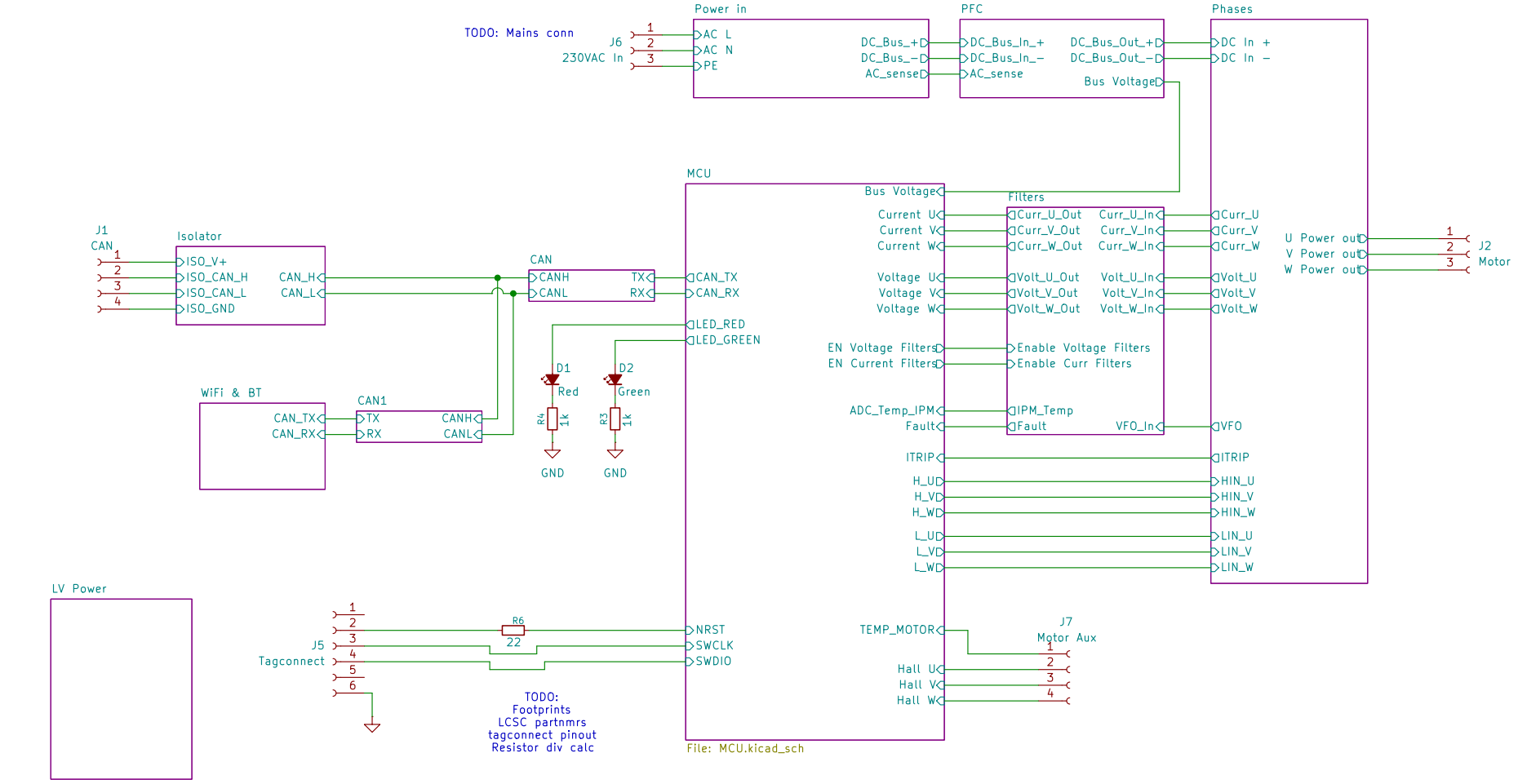
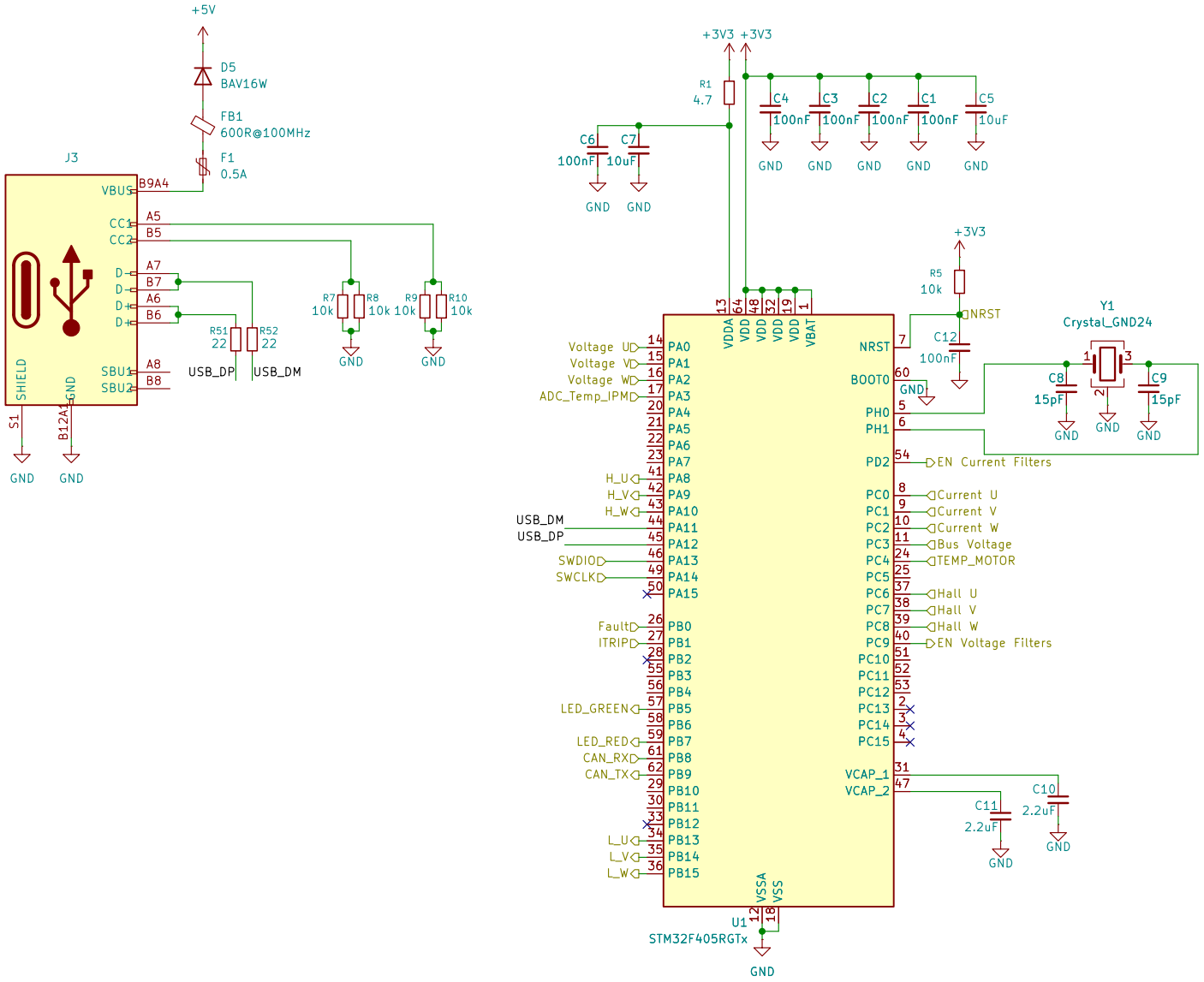


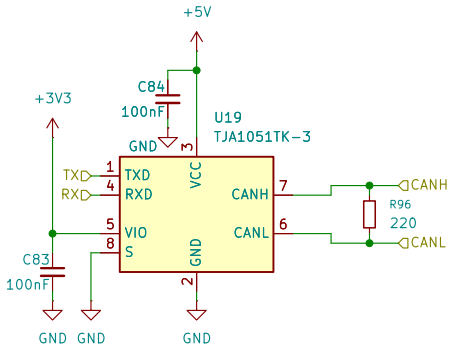
HVAC VESC

Needs to follow for CE cert:
EN IEC 63000 (RoHS)
IEC 60335-2-34 (Particular requirements for motor-compressors)
IEC_61000-3-2 (Limits for harmonic current emissions) <--- Most important

Picobello B.V.	
Sheet: /	
File: CompressorController.kicad_sch	
Title: Mains Powered VESC MD	
Size: A4	Date: 2026-04-16
KiCad E.D.A. 10.0.0	Rev: R1
	Id: 1/11



Picobello B.V.		
Sheet: /MCU/		
File: MCU.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 2/11



Picobello B.V.		
Sheet: /CAN1/		
File: can.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 3/11

DC In + — DC Bus +

DC In − — DC Bus −

+15V

C68 2.2uF — GND

C51 100nF — GND

C52 1nF — GND

U17
IKCM10H60GAXKMA1

VDD 13
VSS 16
NC 24 — GND
VS(U) 1
VB(U) 2
C53 10uF — C54 100nF
VS(V) 3
VB(V) 4
C55 10uF — C56 100nF
VS(W) 5
VB(W) 6
C57 10uF — C58 100nF
HIN(U) 7 ← HIN_UO — R62 ZZ
HIN(V) 8 ← HIN_VO — R63 ZZ
HIN(W) 9 ← HIN_WO — R64 ZZ
LIN(U) 10 ← LIN_UO — R65 ZZ
LIN(V) 11 ← LIN_VO — R66 ZZ
LIN(W) 12 ← LIN_WO — R67 ZZ
VFO 14 — R27 10k — R22 10k — R23 10k — R24 10k — R25 10k — R26 10k — GND
ITRIP 15 — 1k R55 — C60 1nF — GND
P 23 — DC Bus +
U 22 — U Power out
V 21 — V Power out
W 20 — W Power out
NU 19
NV 18
NW 17 — DC Bus −

C59 470nF

IKCM10H60GA
C3615235

U Power out — R36 200k — R37 200k — R38 200k — Volt_U — R39 3.16k — GND

V Power out — R40 200k — R41 200k — R42 200k — Volt_V — R43 3.16k — GND

use 100k here

W Power out — R44 200k — R45 200k — R46 200k — Volt_W — R47 3.16k — GND

TODO Resistor Div calc

+3V3 — C30 100nF — GND

U5
INA240A1D
Curr_U 5
Ref 6
V+ 8
GND 1
Sns1 7
Sns2 3
Out 2
GND

R30 50mR

+3V3 — C33 100nF — GND

U8
INA240A1D
Curr_V 5
Ref 6
V+ 8
GND 1
Sns1 7
Sns2 3
Out 2
GND

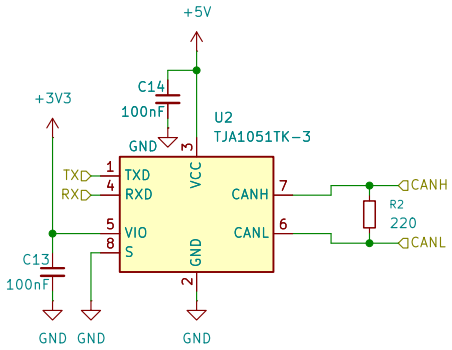
R35 50mR

+3V3 — C35 100nF — GND

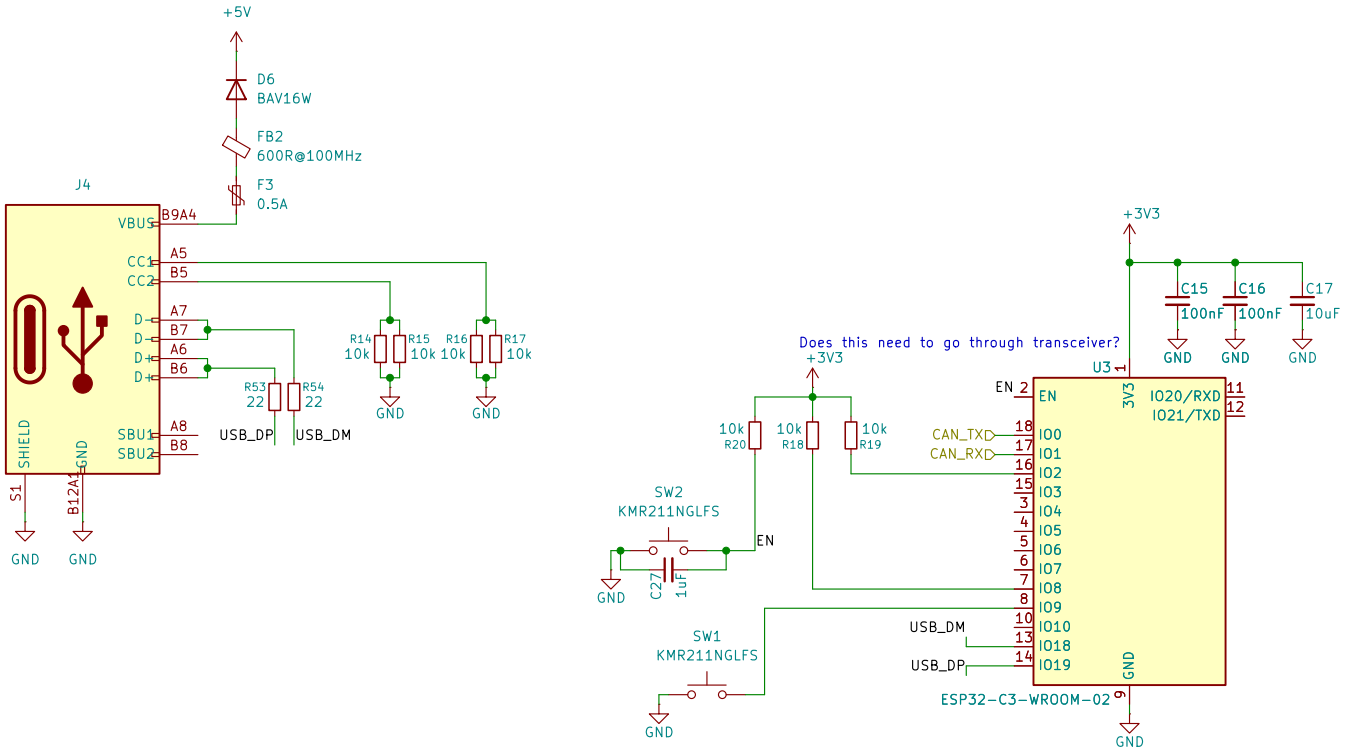
U10
INA240A1D
Curr_W 5
Ref 6
V+ 8
GND 1
Sns1 7
Sns2 3
Out 2
GND

R33 50mR

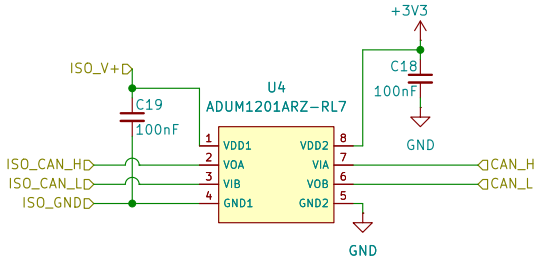
Picobello B.V.		
Sheet: /Phases/		
File: phases.kicad_sch		
Title: Mains Powered VESC MD		
Size: A3	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 4/11



Picobello B.V.		
Sheet: /CAN/		
File: can.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 5/11

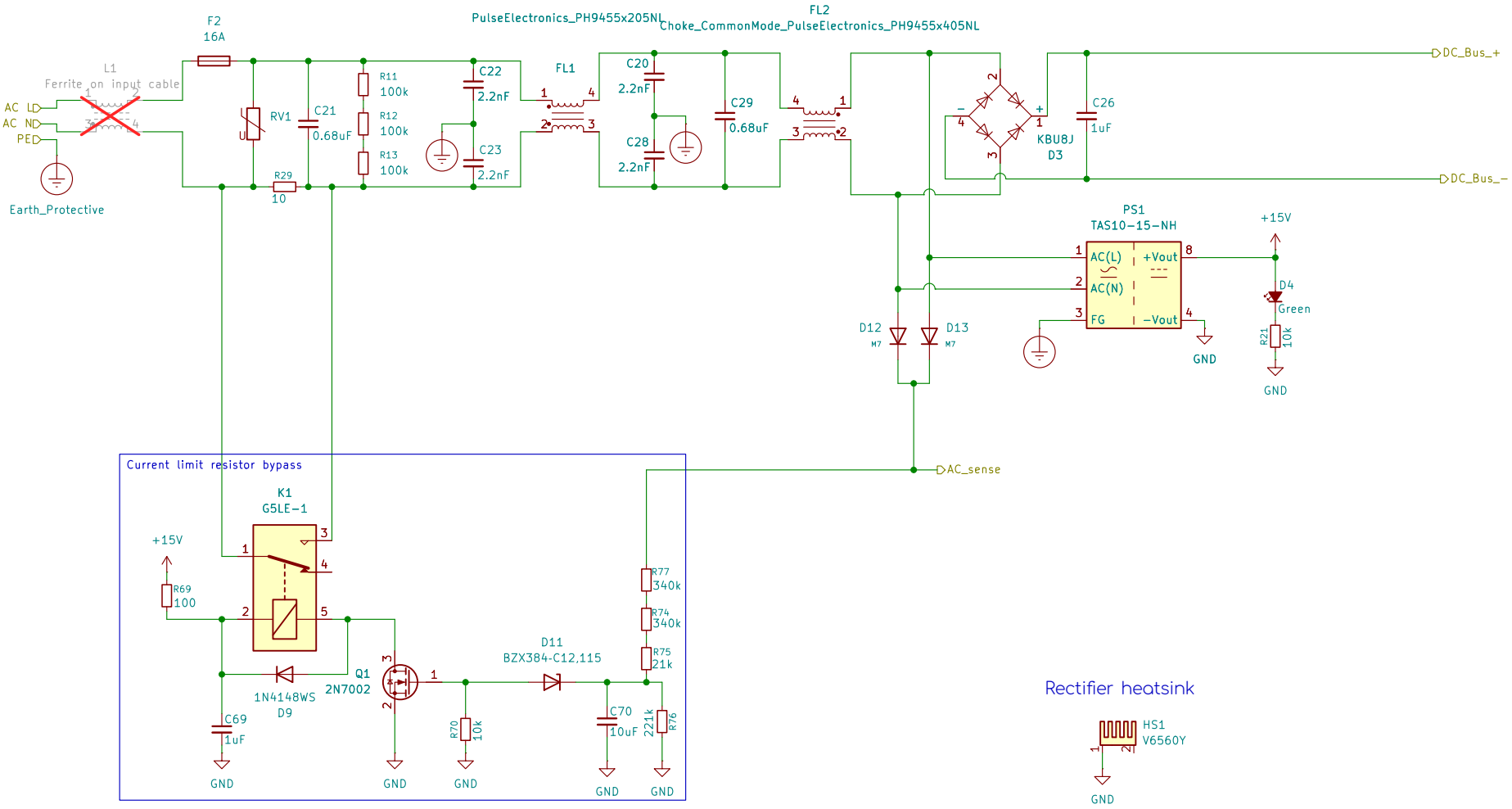


Picobello B.V.		
Sheet: /WiFi & BT/		
File: express.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 6/11

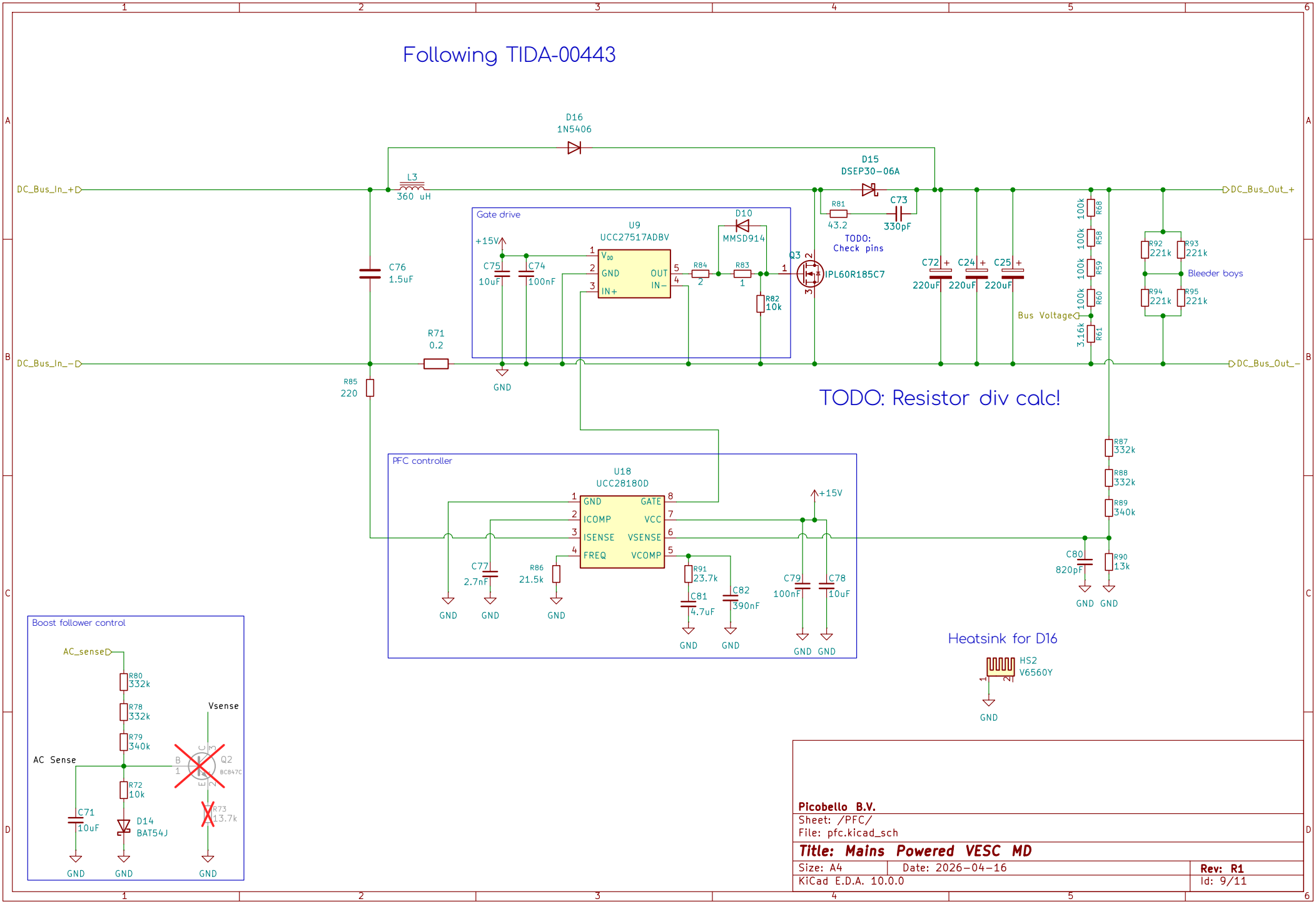


Picobello B.V.		
Sheet: /Isolator/		
File: isolator.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 7/11

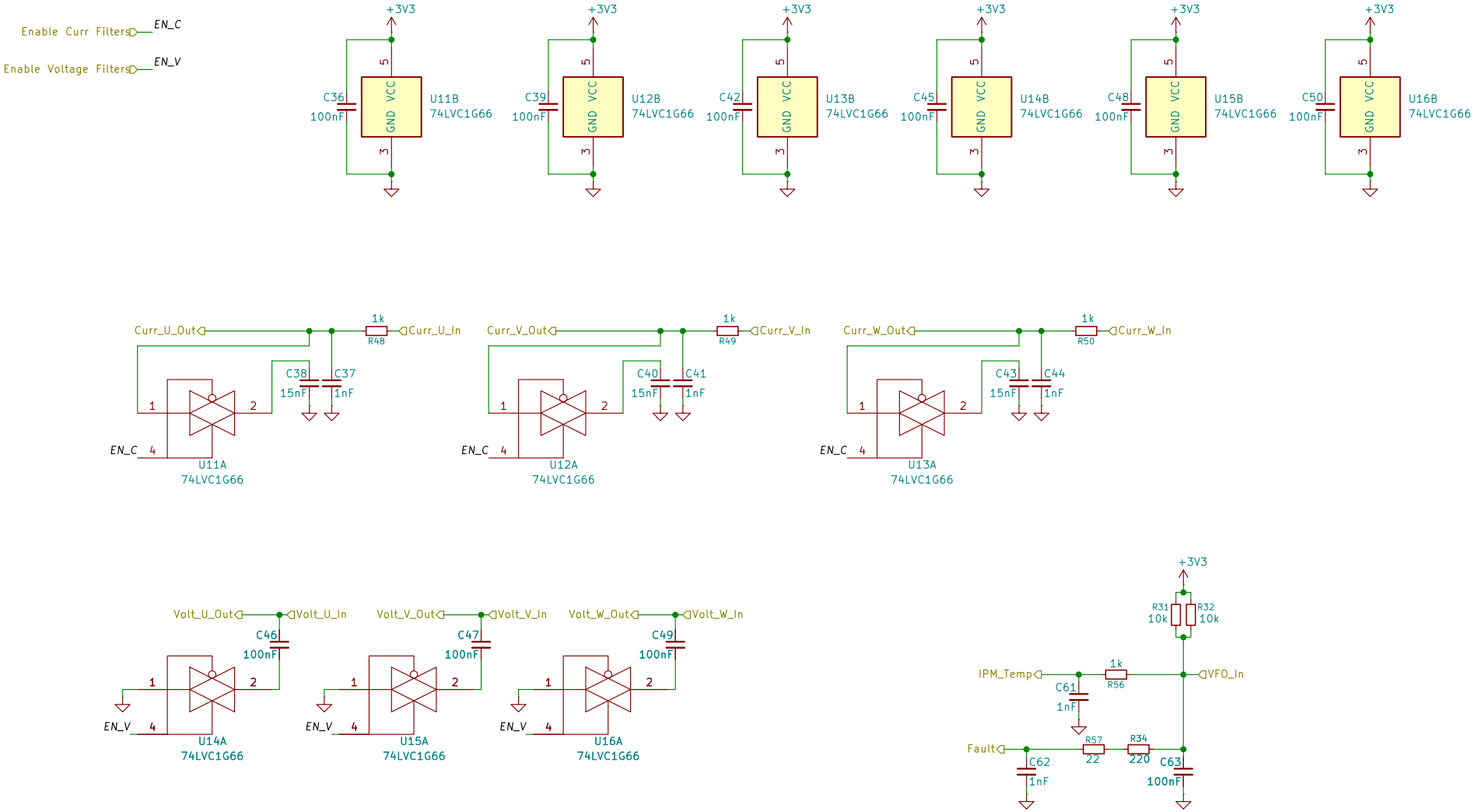
TIDA-00707 Reference design



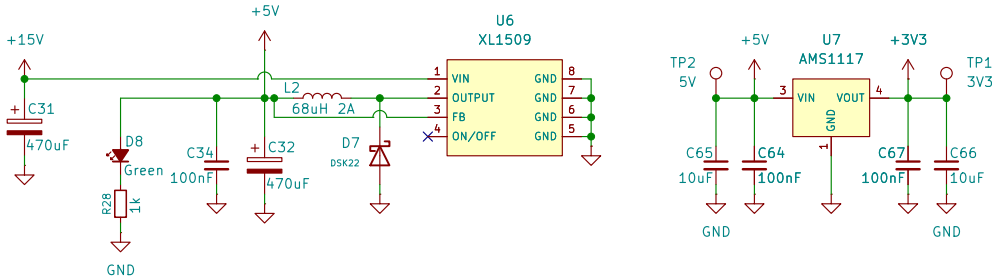
Picobello B.V.		
Sheet: /Power in/		
File: powerin.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 8/11



Picobello B.V.	
Sheet: /PFC/	
File: pfc.kicad_sch	
Title: Mains Powered VESC MD	
Size: A4	Date: 2026-04-16
KiCad E.D.A. 10.0.0	Rev: R1
	Id: 9/11



Picobello B.V.	
Sheet: /Filters/	
File: filters.kicad_sch	
Title: Mains Powered VESC MD	
Size: A4	Date: 2026-04-16
KiCad E.D.A. 10.0.0	Rev: R1 Id: 9/11



Picobello B.V.	
Sheet: /LV Power/	
File: lvpower.kicad_sch	
Title: Mains Powered VESC MD	
Size: A4	Date: 2026-04-16
KiCad E.D.A. 10.0.0	**Rev: R1**
	Id: 10/11